



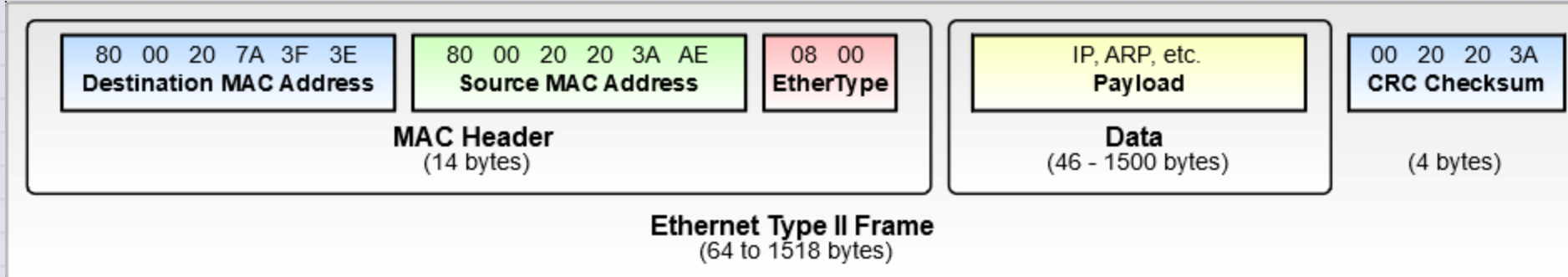
ACIT 2620

Principles of Enterprise Networking

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Ethernet

Frames



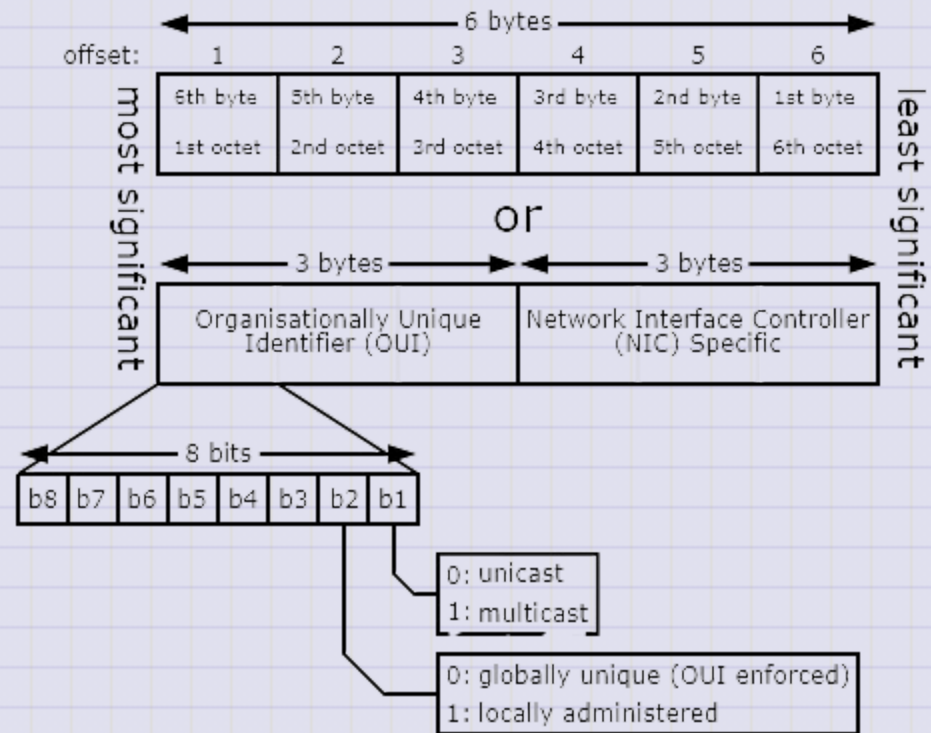
- Preamble
 - Marks beginning of entire frame
- Start of Frame Delimiter (SFD)
 - Indicates beginning of addressing fields
- Destination Address
 - Contains destination node address

- Source Address
 - contains address of sender node
- Length (LEN)
 - indicates length of data/payload
- Data (payload)
 - contains data, or segmented part of data, transmitted from originating node

- Pad
 - Used to increase size of the frame to its minimum size requirement of 46 bytes
- Frame Check Sequence
 - provides algorithm to determine whether data were correctly received
 - most commonly used algorithm is Cyclic Redundancy Check (CRC)

Ethernet Addressing

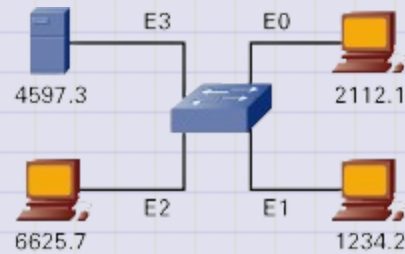
- MAC address: Media Access Control (MAC) sub-layer
- 48 Bits
- Number uniquely defining a network node
- Generally rendered as Hex: **00:1e:33:ba:87:c1**
- Doesn't contain any data regarding network location –just an ID



- first three bytes
 - Either Manufacturer hard coded
 - Or Reserved Addresses (common ones)
 - Broadcast Address **FF:FF:FF:FF:FF:FF**
 - Spanning Tree Multicast: **01:80:C2:00:00:00**
 - IANA reserves all address starting with **00:00:5E** see Ethernet Numbers(this includes IPv4 multicast -and inserts the low 23 Bits of the multicast IPv4 Address into the Ethernet Address)
 - **33:33:XX** is reserved for IPv6 Multicast

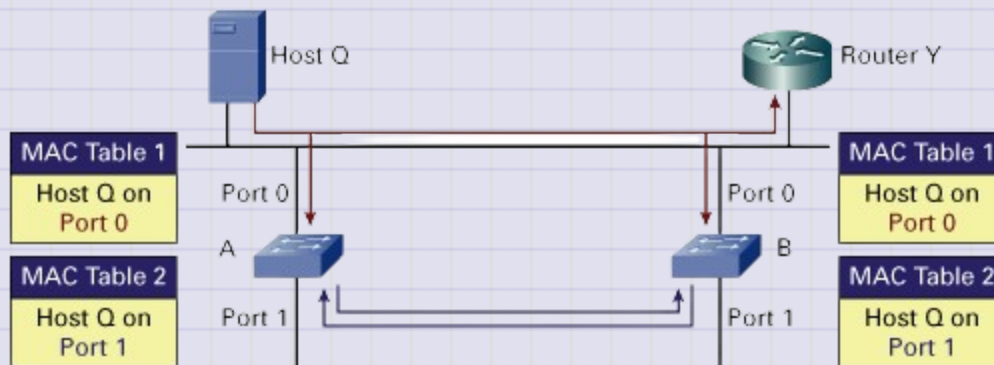
Switching

- Making Forwarding decisions
 - Transparent bridging

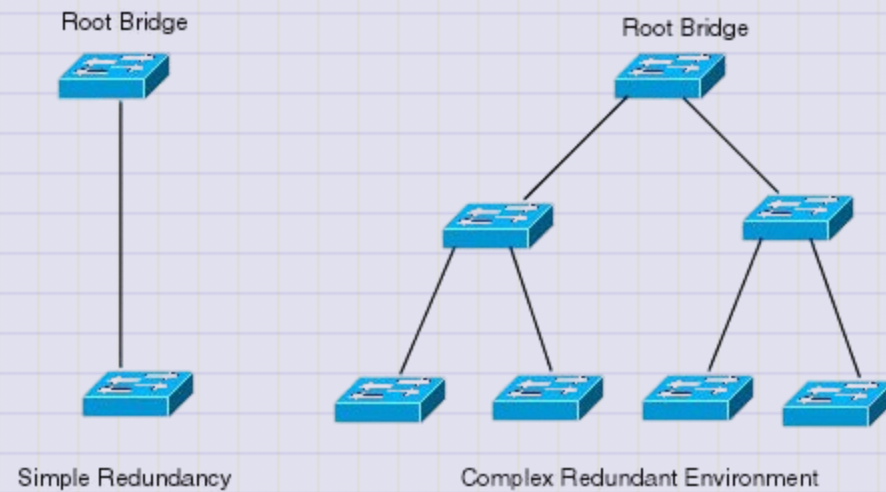


Time 0		Time 1		Time 2	
Port	Address	Port	Address	Port	Address
		E2	6625.7	E2	6625.7
				E0	2112.1

Broadcast Loop and STP

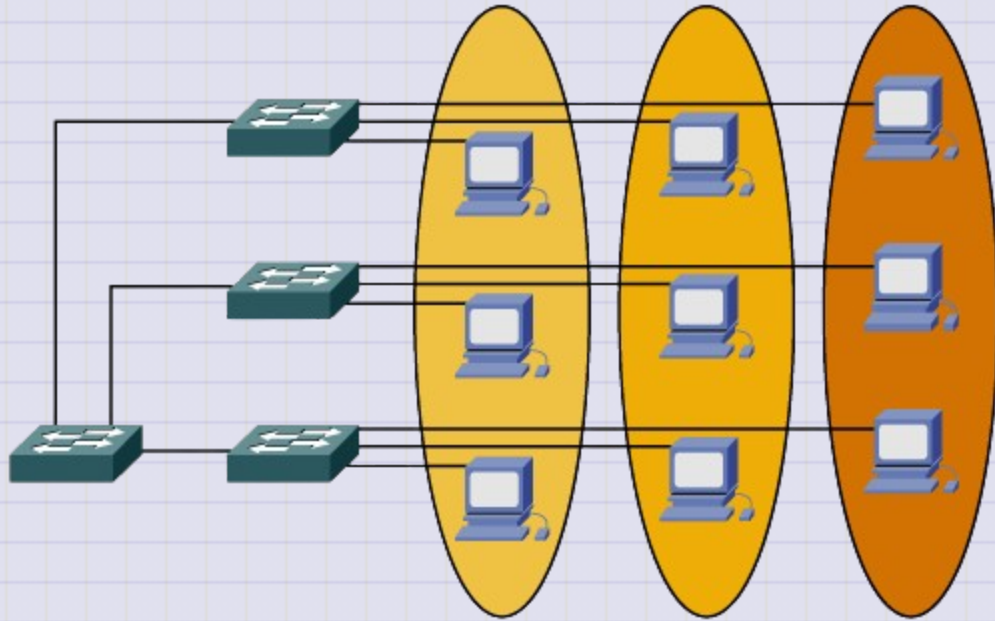


Spanning Tree Protocol



VLANs

- Virtual Local Area Networks
- A logical network within a physical network
- Achieved by grouping some of the switch ethernet ports into a logical broadcast domain
- Can span multiple switches

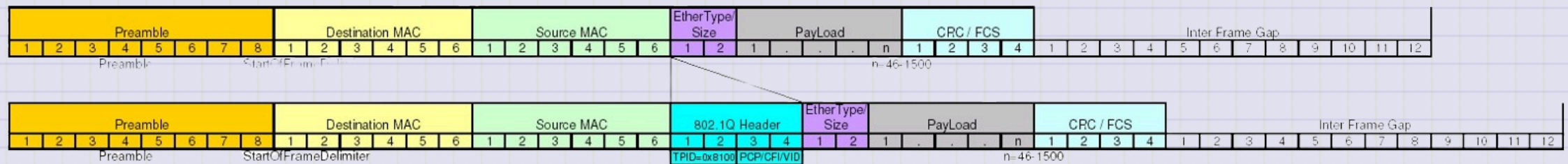


VLAN port types

- Access ports
 - assigned VLAN ID
 - for connecting end hosts/nodes
 - nodes connected to ports with same VLAN ID are in the same broadcast domain

- Trunk ports
 - typically for switch to switch or switch to router connection
 - carry "tagged" frames, i.e modified ethernet frames with VLAN markers

Tagged frames



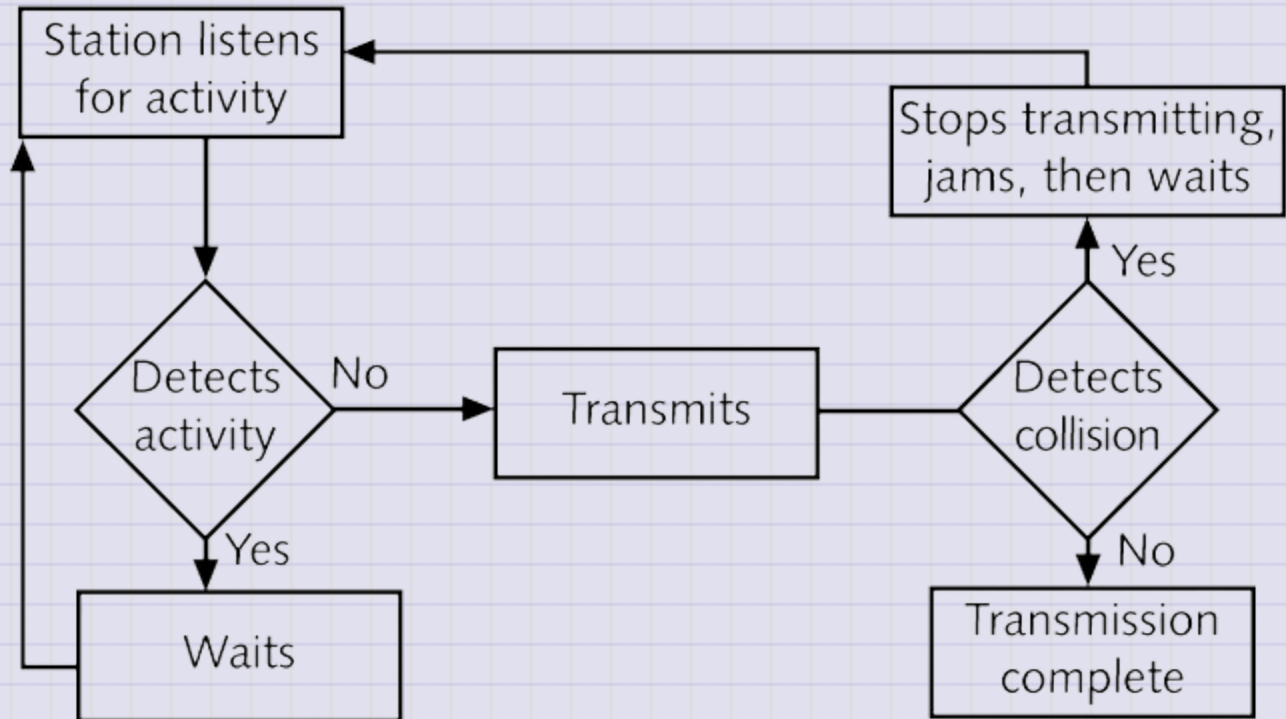
- 4-byte tag header inserted between Source MAC and EtherType fields
 - 2-byte tag protocol identifier (TPID)
 - a fixed value of 0x8100 that indicates the frame carries tag information.

- 2-byte tag control information (TCI)
 - Three-bit user priority (used to prioritize traffic)
 - Drop Eligible Indicator (DEI) (in congestion is frame “dropable”)
 - Twelve-bit VLAN identifier (VID)-Uniquely identifies the VLAN to which the frame belongs

Link Access Methods

- Manage shared medium access contention (collision)
- Two methods:
 - CSMA/CD: for wired Ethernet
 - CSMA/CA: for wireless Ethernet

CSMA/CD



CSMA/CA

